Mock Midterm Examination

Machine Learning in Engineering Science

Fall 2026 | Revised 9 November 2026 | NCKU + LAiMM

Practice candidate paper |75-minute written practice |100 raw points

Name _____ Student ID _____

One A4 sheet handwritten on both sides and a standard calculator are permitted. During timed work, do not use the internet, AI or execute code.

Answer all six questions on the numbered sheets. Use labels, short bullets, fractions and equations. All signed residuals mean prediction minus observation. Suggested answer lengths guide pacing, not grading. The stated question times include reading and answering; reserve the final four minutes for checking.

Activity	Points	Minutes
Read and plan	0	3
1 Frame a vibration prediction experiment	20	12
2 Evaluate alarms under asymmetric costs	15	10
3 Relate variance retention to prediction	10	7
4 Reason about a causal sequence model	25	19
5 Diagnose variation and a changed deployment	20	13
6 Check predictions before claiming reproduction	10	7
Final check	0	4
Total	100	75

This ungraded Mock covers Weeks 1–10 core, excluding holidays and optional challenges. Question 6 is a diagnostic before today's workflow Lab; revisit its reasoning after the lesson. The formal midterm assesses actually taught content and remains written plus individual two-minute oral, with no midterm project. Today's separate 15-minute review starts after the 75-minute written attempt.

Question 1 Frame a vibration prediction experiment

20 points | Suggested time 12 minutes

A pump dataset contains repeated runs from 18 pumps. At the start of a run, load R in percent, rotation speed U in thousands of revolutions per minute, and ambient temperature are known. Predict vibration RMS ten seconds later in mm/s. Deployment concerns previously unseen pumps. X contains only these available inputs; pump IDs are stored separately.

```
Z = StandardScaler().fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(
    Z, y, test_size=0.25, random_state=12)
```

(a) 4 points State the observational unit, target, prediction time and type of supervised task.

(b) 6 points Give two problem→repair bullets: (i) pump membership across splits and (ii) preprocessing scope. Include the role of validation and when the final comparison is frozen; no library syntax is required.

(c) 4 points A separate RAW-input model predicts RMS = 0.5 + 0.05R + 0.2U. For R = 30, U = 2 and observed RMS = 3, calculate the prediction, signed residual (prediction minus observation), absolute error and squared error.

(d) 6 points Name a training-only constant baseline suited to RMSE; state one fair-comparison rule; then give one Ridge effect and one evaluation problem Ridge cannot repair.

Answer format: Use four labels in a; two problem→repair bullets in b; four values in c; three short bullets in d. About 70 prose words plus calculations are enough.

Question 2 Evaluate alarms under asymmetric costs

15 points | Suggested time 10 minutes

For a–b, use the TEST table at threshold 0.7; positive means a faulty pump. Part c uses a DIFFERENT VALIDATION set. Its 0.7 counts happen to match the test counts. Total cost is C =8FN + FP. The test table contains no predictions for threshold 0.4.

Test actual class	Predicted faulty	Predicted healthy
Faulty	12	8
Healthy	6	54

Validation threshold	TP	FN	FP	TN
0.4	18	2	20	40
0.7	12	8	6	54

(a) 4 points Identify TP, FN, FP and TN in the TEST table.

(b) 6 points Calculate precision, recall (sensitivity), specificity and F1. Fractions or three-significant-figure decimals are acceptable. Use F1 = 2TP / (2TP + FP + FN).

(c) 5 points Compute validation cost at each threshold and choose the lower-cost setting. State when to freeze it. Can this test table give the chosen setting's final test score? Give one reason.

Answer format: Use four counts in a, four fractions in b, then the two costs and two short decision statements in c.

Question 3 Relate variance retention to prediction

10 points | Suggested time 7 minutes

Two standardized pump features have covariance [[1, 0.6], [0.6, 1]]. The supplied eigenvalues are 1.6 and 0.4. Unit eigenvectors are shown below. Consider z = (2, 0). Scores are dot products; PC1-only reconstruction is score 1 times v1.

$$v_1 = \frac{(1,\ 1)}{\sqrt{2}} \quad v_2 = \frac{(1,\ -1)}{\sqrt{2}}$$

(a) 2 points Calculate PC1's retained-variance fraction.

(b) 2 points Calculate z's PC1 score and reconstruction using PC1 alone.

(c) 2 points A fault is identified by the difference between the standardized features. Explain what PC1-only reconstruction can remove, even if it retains most variance.

(d) 4 points In a SEPARATE K-means fit on raw coordinates, what changes if one coordinate is converted from seconds to milliseconds without rescaling? Why do cluster IDs alone not identify physical fault modes?

Answer format: Use one fraction, one score and one reconstruction; then three short mechanism statements. About 35 prose words plus calculations are enough.

Question 4 Reason about a causal sequence model

25 points | Suggested time 19 minutes

Use the LSTM equations below. All gate outputs and the candidate are supplied. Attention logits are already scaled, immediately before masking and softmax.

$$c_t = f_t c_{t-1} + i_t g_t$$

$$h_t = o_t \tanh(c_t)$$

(a) 4 points For x0=1,x1=3,x2=2,x3=5,x4=4, use a four-value lookback at time 3 to predict x4. Give input and target, why x4 is unavailable, and one chronological split rule.

(b) 6 points Given c_previous = 1, f_t = 0.6, i_t = 0.5, g_t = 0.8 and o_t = 0.25, calculate c_t and h_t using tanh(1) = 0.7616. Then set ONLY o_t to zero and explain what happens to the hidden output and the stored cell state.

(c) 8 points Scaled logits are [ln(2),0,ln(3)] with exponentials [2, 1, 3]. Key 2 is padding; keys are numbered 1–3. V1=[5, 0],V2=[100, 100],V3=[0, 5]. Give masked normalized weights and context. Explain why only zeroing the padded weight AFTER unmasked softmax is insufficient.

(d) 7 points A fixed deterministic causal decoder receives two length-10 inputs with identical positions 0–3 and changed positions 4–9; position indices and all other inputs are unchanged. Answer three rows: (i) which logits must agree, and why; (ii) why teacher-forced training can be parallel but ordinary generation is sequential; (iii) what explicit positional information adds beyond the causal mask.

Answer format: Use the requested values and short labeled statements. In d answer three rows: prefix check, training/generation, positions. About 70 prose words plus calculations are enough.

Question 5 Diagnose variation and a changed deployment

20 points | Suggested time 13 minutes

At one fixed input, the known noise-free target is 2.0. The controlled synthetic study has independent zero-mean observation-noise variance 0.25. The table summarizes 20 independent training resamples. Treat its reported moments as the supplied estimates. Use expected MSE = squared bias + refit variance + noise variance; do not take a square root.

Model	Training rows	Mean prediction	Refit variance
Affine	80	1.6	0.25
Affine	800	1.6	0.01
Nonlinear	800	2.0	0.08

(a) 6 points Calculate squared bias and expected MSE for the 800-row affine model, and expected MSE for the 800-row nonlinear model.

(b) 4 points Use the two affine rows to state what more data changed and what it did not. Explain one possible cause of a persistent curved mean residual and what happens to the stated observation-noise floor.

(c) 4 points Distinguish A, deployment humidity values become more frequent in a different range with the same conditional response, from B, pump ageing changes the conditional response at the same inputs.

(d) 6 points A frozen model has MAE 3 on 20 high-load cases and MAE 0.5 on 80 other test cases. Compute total test MAE and its reweighted value when high-load cases become 40 percent of deployment, assuming within-group mean absolute errors stay fixed. Explain the assumption and one limitation.

Answer format: Show the short calculations; answer b–d in compact bullets. About 65 prose words plus calculations are enough.

Question 6 Check predictions before claiming reproduction

10 points | Suggested time 7 minutes

A fresh receiver process refits the frozen model pair. The following row-order difference occurs for both baseline and candidate test predictions. The required input contract declares rotation speed in revolutions per minute, abbreviated rpm.

Reference ID	Prediction	Receiver ID	Prediction
9	3	7	1
4	8	9	3
7	1	4	8

(a) 3 points State the required ID checks, how to align the files, and the aligned maximum absolute prediction difference.

(b) 2 points Why is matching a rounded MAE inadequate compared with aligned vector comparison for both models at a predeclared tolerance?

(c) 2 points A copied contract changes rpm to revolutions per second while leaving values unchanged. What must an rpm-only loader do before fitting, and why?

(d) 3 points Give one claim supported by matching hashes/replay, one stronger claim they do not establish, and one environment version to record.

Answer format: Use an ID mapping and zero-difference result in a, then one short statement per requested claim. About 35 prose words plus calculations are enough.

Written examination answer sheet

Student ID _____ Label each part a b c or d clearly.

Question 1

Question 2

Written examination answer sheet

Student ID _____ Label each part a b c or d clearly.

Question 3

Question 4

Written examination answer sheet

Student ID _____ Label each part a b c or d clearly.

Question 5

Question 6

Two minute oral practice

Use one case per two-minute slot. Read only its context and main question during the first 15 seconds; keep the follow-up covered until asked. Give a brief decision, supplied evidence, next step and limitation. The follow-up's30 seconds includes examiner speech. Use the separate worked-solutions guide for scoring. Do not grade accent or verbosity as ML understanding.

M1 Selection after test

A student tries five Ridge alphas, selects the lowest test RMSE, and reports it as an unbiased final estimate.

Main question: Identify the problem. Use the evidence to propose a next step and state one limitation.

Follow-up, shown after the main response: Would a fixed random seed make that estimate valid? Why?

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M2 A unit label in a replay

An airflow package requires m/s. Its copied contract says km/h with unchanged numbers. The loader rejects it before fitting.

Main question: Is this useful rejection or model failure? Give an evidence-based next step and one limitation.

Follow-up, shown after the main response: Does relabeling it to m/s prove the original units were correct? Why?

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Pacing:15 seconds reading +60 main response +30follow-up including examiner speech +15 closure =120 seconds. This is practice, not a whole-class oral examination inside the review period.